

Hiperbolic Structures and Triangulations: Gluing and Completeness Equations for the Borromean Rings

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1 Introduction

This article was written to facilitate the reader into learning some basic concepts of knot theory by focusing on one type of problem that can be encountered in the field. The paper was written in such a way that anyone with a certain degree of interest in mathematics coming from a high school education might understand.

The process gone through yielded some diagrams that some readers could find beautiful and might incentivize them into further investigation and take a deeper dive into the field, in such case it is highly encouraged to start with Jessica Purcell's excellent paper *Hiperbolic Knot Theory* in order to grasp deeper concepts that might have been overlooked by this paper in order to keep things as accessible as possible.

As for any field, having more researchers would be great as it will entail more knowledge and discoveries that in the future might aid in other fields in science by giving them more tools at their disposal.

1.1 Objective

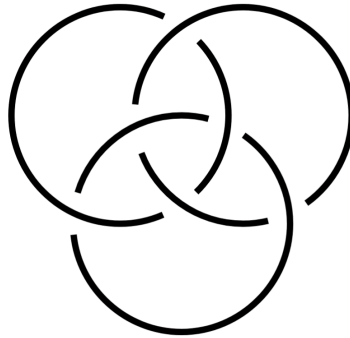


Figure 1: Diagram of the borromean rings [[mathworldBorr](#)].

The main goal of this article is to find the gluing and completeness equation of the borromean rings which is shown in Fig[1]. In order to find these equations, the tetrahedra decomposition of the borromean ring complement will be studied to generate a hyperbolic structure proposal for the three components of the Borromean rings. By finding out how each of these tetrahedra will be glued in order to get a 3-manifold (a 3 dimensional space, more specifically a 3 dimensional hiperbolic space) that is homeomorphic to the borromean ring complement the gluing the equations can be found.

1.2 Definitions

Definition 1.1 (Knot). A closed 3-dimensional surface with crossings. A simple loop with no crossings is known as a *unknot*. The most basic knot with at least one crossing is known as a trefoil knot that is shown in Figure[2]. The trefoil knot is also the result when first crossing and pulling the string and joining the ends of the string when tying shoelaces.

Definition 1.2 (Link). When multiple knots are entangled, it is known as a link. The borromean rings shown on Fig[1] is a link between three unknots. This can be seen by following the string or curve from a starting

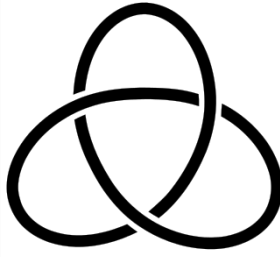


Figure 2: The trefoil knot, the most basic knot after the unknot [trefoil].

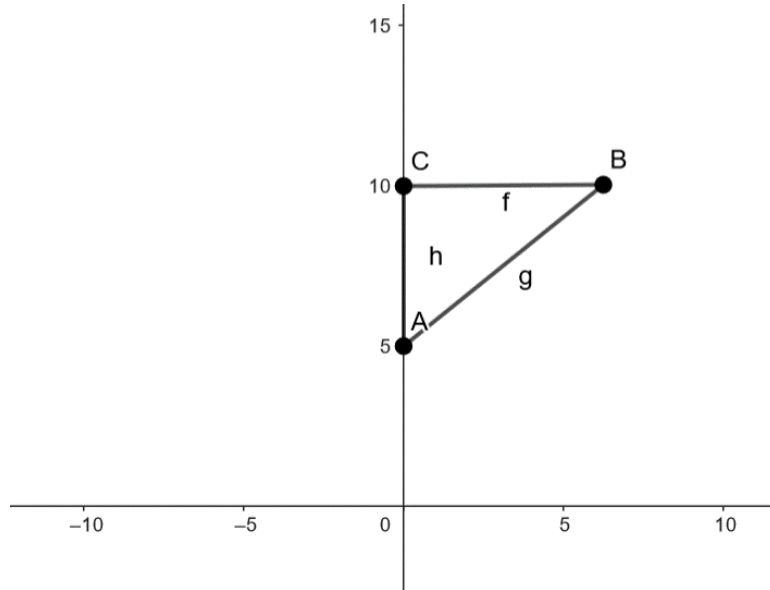


Figure 3: In the euclidean plane, geodesics are always straight line segments.

point; by following it, the path taken did not cover the whole link since the curve only covered one of three knots in the link.

Definition 1.3 (Knot/Link complement). It is everything but the knot/link in question. In both Fig[1] and Fig[2], the complement is the space covered in white (knots and links exist in 3 dimensions or higher). This is similar to the complements in probability: if the probability of an event happening is 12%, then the complement of that probability is all possibilities that exclude that given event, in this case it would be 88%.

Definition 1.4 (Hiperbolic plane). The hiperbolic plane is a 2-dimensional space, commonly written as $\mathbf{H}^2$, whose metric is defined as:

$$ds^2 = \frac{dx^2 + dy^2}{y^2} \quad (1)$$

For reference, the commonly used euclidean metric is defined as:

$$ds^2 = dx^2 + dy^2 \quad (2)$$

By comparing both metrics, the hyperbolic metric in equation 1 is similar to the euclidean metric in equation 2, but compressed by a factor of y . Points on the hyperbolic plane can be represented on the upper complex plane, that is all complex numbers $z = x + yi$ where $y > 0$. The distance between two euclidean points can be calculated through a straight line whose end points are the two points of interest, this straight line that represents the shortest distance between two points is commonly referred to as the *geodesic*. In the hyperbolic plane however, the shortest distance between two points is not always calculated with straight lines, arcs play an important role in calculating this distance as well.

In Fig[3], notice that to calculate the distance from any points A, B and C, a straight line is always the shortest distance (or geodesic) from any pair of these points. Similarly, in Fig[4], if two points are vertical from one another (A and B or H and I), then the shortest distance from these two points is a straight vertical line. However, if two points are not vertically aligned (points F and G or C and D) then the geodesic becomes an arc that connects these two points and whose center is located on the x -axis.

To go more thoroughly on the geodesics shown on Fig[4]: points A and B are have the same x -component,

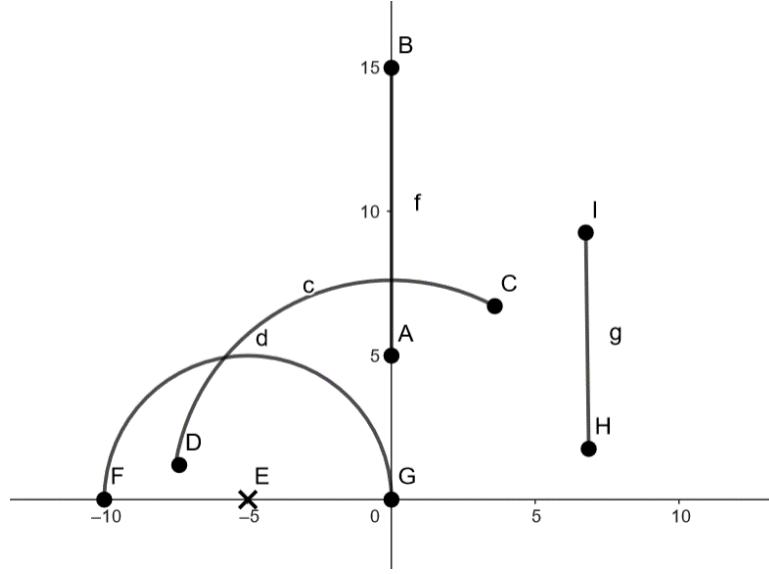


Figure 4: In the hyperbolic plane, geodesics can be either vertical line segments or arcs.

thus they are vertically aligned and thus the vertical line segment is the geodesic of those two points. The same can be said for points H and I. Points G and F lie in the x -axis, thus the arc connecting these two points lie in the midpoint, that is on $z = -5$ (marked as E). Points C and D are not vertically aligned, thus an arc with center at the x -axis (in this case, the center is G) will be the geodesic.

Definition 1.5 (Hyperbolic space). Similar to the hyperbolic plane, hyperbolic space is taking the same metric but in 3 dimensions. Now a point in hyperbolic space is seen as $(x + yi, t)$, where $y > 0$ and $t > 0$. Again, geodesics are straight vertical lines or arcs centered at the x -axis

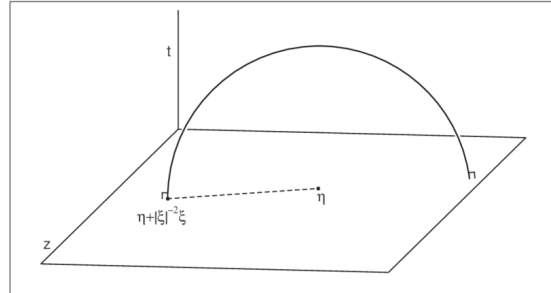


Figure 5: Upper half plane model for the hyperbolic space [hypSpaceDiagram]

Definition 1.6 (Möbius Transformation). Is a transformation, $f(z)$ in $\mathbb{C}$ that takes a complex number, z and returns another complex number in the form:

$$f(z) = \frac{az + b}{cz + d}$$

Where $a, b, c, d \in \mathbb{C}^4, ad - bc \neq 0$ [mobius] Möbius Transformations has some interesting properties such as bijectivness (for every value of z there is a unique $f(z)$, and for every $f(z)$ there is a unique z , in other words there is an inverse transformation of f), and line perservation (transforms lines to lines and circles to circles) that will become helpful later when talking about *Edge Invariants*

Definition 1.7 (Ideal Tetrahedra). Before defining an ideal tetrahedron, remember that a tetrahedron is the most basic polyhedron in 3-dimensional space, with 4 vertices, 6 edges and 4 faces. Euclidean tetrahedra can be thought as triangular pyramids. An ideal tetrahedron in hyperbolic space is a tetrahedron in which all vertices lie in the boundary of the space. The boundary $\partial\mathbf{H}^3$ consists of the plane $y = 0$, the complex plane and the point at ∞ . Since some of these points could end up being excluded from the hyperbolic space (they could be in an excluded boundary like the x -axis or real axis), ideal tetrahedra will have all of their vertices removed.

Claim 1.8 (Edge invariants). *Before talking about edge invariants, it is crucial to understand that all ideal tetrahedra can be transformed to an ideal tetrahedron with edges at $0, 1, \infty$ and some z . To prove this, assume*

that an ideal tetrahedron has vertices at z_1, z_2, z_3 and z_4 , then let's consider a Möbius transformation, f , that takes $z_1 \rightarrow 0$, $z_2 \rightarrow 1$, $z_3 \rightarrow \infty$ and $z_4 \rightarrow z$ (in order, conditions 1 through 4). Then:

$$\begin{aligned} f(t) &= \frac{a(t - z_1)}{t - z_3} \quad (\text{From conditions 1 and 3}) \\ f(z_2) &= 1 = \frac{a(z_2 - z_1)}{z_2 - z_3} \quad (\text{Condition 2}) \\ a &= \frac{z_2 - z_3}{z_2 - z_1} \\ f(t) &= \frac{z_2 - z_3}{z_2 - z_1} \frac{t - z_1}{t - z_3} \end{aligned}$$

As a last step, define the result of $f(z_4)$ as z to satisfy condition 4 and thus all ideal tetrahedra in hyperbolic space can be transformed in such a way that three vertices lie in $0, 1$ and ∞ and the fourth lies in some complex number Z . Thus for every z there is a unique tetrahedron relating to it. When dealing with multiple tetrahedra and gluing them together on different orientations, the value of z for each tetrahedron might change. The equations giving insight into the change of z are called the edge invariants and are based on what edge of the tetrahedron is being focused on.

Define z_{01} as the edge invariant of edge with vertices 0 and 1 , z_{02} with the edge with vertices at 0 and ∞ , z_{03} with edge 0 and z , z_{12} with 1 and ∞ and z_{23} with ∞ and z . Define $z_{01} = z$

If dealing with the edge with vertices at ∞ and z , one must find the Möbius transformation that fixes sends 0 to 1 , 1 to 0 and z to ∞ . This transformation would end up being:

$$w \rightarrow \frac{z(w - 1)}{w - z}$$

Notice that the transformation sends ∞ to z , thus the first edge invariant equation is:

$$z(e'_1) = z(e_1) \quad (3)$$

Here, e_1 represents the edge with vertices at 0 and 1 and e'_1 represents the opposite edge (the edge with vertices at Z and ∞). This equation can also be applied to any pair of opposite edges, that is the value of $z(e'_n) = z(e_n)$, thus $z_{01} = z_{23}$, $z_{02} = z_{13}$ and $z_{03} = z_{12}$

For the edge with vertices at 1 and ∞ , the edge invariant can be obtained from the transformation that fixes ∞ , takes 1 to 0 and takes z to 1 . This transformation is:

$$w \rightarrow \frac{w - 1}{z - 1}$$

And takes 0 to $-\frac{1}{z-1}$, thus the second edge invariant is:

$$z(e_2) = -\frac{1}{z - 1} \quad (4)$$

Here e_2 is the edge with vertices at 1 and ∞ , hence $z_{12} = z_{03} = -\frac{1}{z-1} = \frac{1}{1-z}$.

For the edge with vertices at ∞ and z , the edge invariant can be obtained from the transformation that fixes ∞ , takes z to 0 and takes 0 to 1 . This transformation is:

$$w \rightarrow -\frac{w - z}{z}$$

And takes 1 to $-\frac{1-z}{z}$, thus the third and last edge invariant is:

$$z(e_3) = \frac{z - 1}{z} \quad (5)$$

Here e_3 is the edge with vertices at 0 and ∞ , hence $z_{12} = z_{03} = \frac{z-1}{z}$.

Theorem 1.9 (Edge gluing equations). *Let M^3 admit a topological ideal triangulation such that each ideal tetrahedron has a hyperbolic structure. The hyperbolic structures on the ideal tetrahedra induce a hyperbolic structure on the gluing, M , if and only if for each edge e ,*

$$\prod z(e_j) = 1 \text{ and } \sum \arg(z(e_j)) = 2\pi,$$

where the product and sum are over all edges that glue to e [Purcell].

Definition 1.10 (Cusp torus). Let M be a 3-manifold with a torus boundary. Define a *cuspidal neighborhood* of M to be a neighborhood of ∂M homeomorphic to the product of a torus and an interval, $T^2 \times I$. Define a *cuspidal torus* to be a torus component of ∂M , or the boundary of a cuspidal neighborhood [Purcell].

Theorem 1.11 (Complete structure). *Let M be a 3-manifold with a torus boundary and hyperbolic structure. Then the structure on M is complete if and only if for each cusp of M , the induced structure on the boundary of the cusp is a Euclidean structure on the torus [Purcell].*

Definition 1.12 (Cusp triangulation). Let M have a topological ideal triangulation. If we truncate the vertices of each ideal tetrahedron, we obtain a collection of triangles, each of which lies on the boundary of a cusp. Edges of each triangle inherit a gluing from the gluing of faces of the ideal tetrahedra. This gives a triangulation of each boundary torus, which we call *cuspidal triangulation* [Purcell].

Definition 1.13 ($H(\alpha)$). Suppose M has a topological ideal triangulation, and let T be the boundary torus of a cusp of M . Let $\alpha \in \pi_1(T)$, so α is a loop on T in the homotopy class of α . We associate a complex number $H(\alpha)$ to α as follows. First, orient the loop α on T . The loop α can be homotoped to run through any triangle of the cuspidal triangulation of T monotonically, i.e. in such a way that it cuts off a single corner of each triangle it enters. Denote the edge invariants of the corners cut off by α by $z_1, z_2, \dots, z_n$. Further associate to each corner a value $\epsilon_j = \{-1, 1\}$: if the j -th corner cut off by α lies to the left of α , set $\epsilon_j = 1$. If corner lies to the right of α , set $\epsilon_j = -1$. Finally, set the value of $H(\alpha)$ to be [Purcell]:

$$H(\alpha) = \prod_{j=1}^n z_j^{\epsilon_j}.$$

Claim 1.14 (Completeness equations). *Let T be the torus boundary of a cuspidal neighborhood of M , where M admits a topological ideal triangulation, and the ideal tetrahedra admit hyperbolic structures that satisfy the edge gluing equations. Let α and β generate $\pi_1(T)$. If $H(\alpha) = H(\beta) = 1$, then the ideal triangulation is a geometric ideal triangulation, i.e. the hyperbolic structure on M induced by the hyperbolic structure on the tetrahedra will be a complete structure. The equations $H(\alpha) = 1$ and $H(\beta) = 1$ are called the completeness equations [Purcell].*

2 Methodology

2.1 Topological ideal triangulation of the knot complement

Remark 2.1 (Regina). Regina is a software package for 3-manifold and 4-manifold topologists, focusing on triangulations, knots and links, among others [Regina]. For 3-manifold, Regina provides information on known 3-manifolds for easy access, such as the case study of interest: the Borromean link.

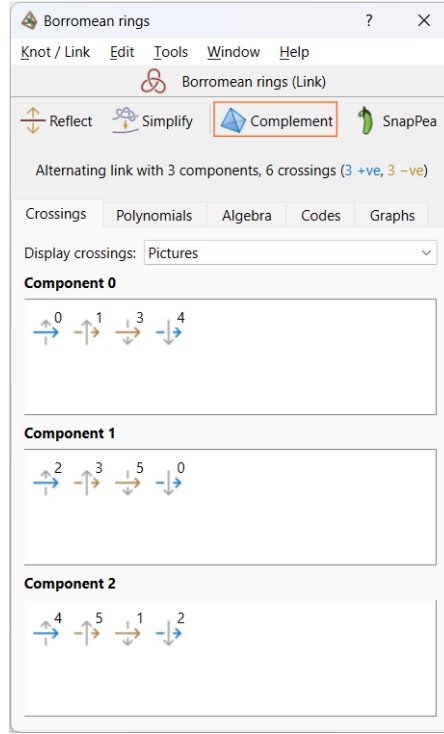


Figure 6: Borrmean Link Description.

Once you have accessed Regina and selected the Borrmean link, you can obtain detailed information from this link. In Fig[6] we see how the 3 nodes that make up the link behave and, more importantly, we can obtain information about the link's complement.

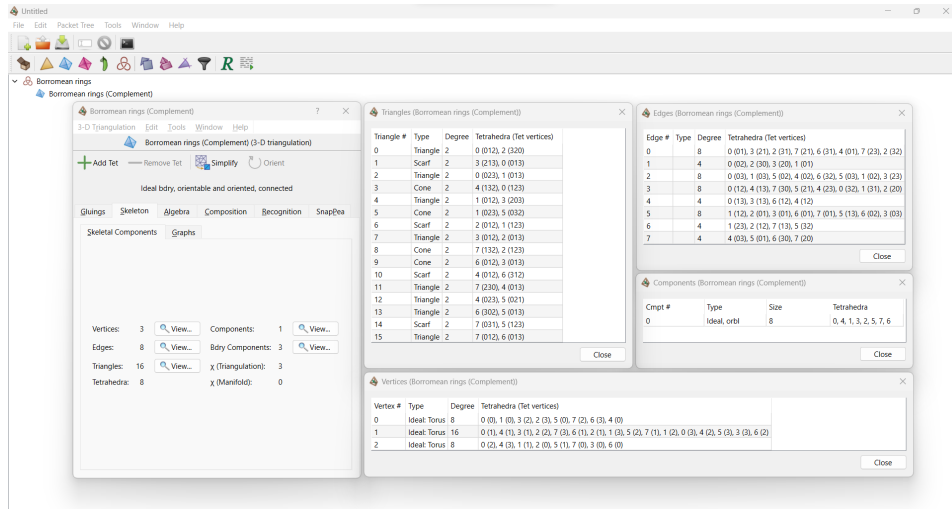


Figure 7: General visualization of the skeleton of the Borrmean link complement.

Initially, it shows what the pastes look like for the link plugin, but we will focus on the skeleton tab, which shows in detail how the plugin triangulation will work. As seen in Fig[7], everything necessary to obtain the equations of interest appears. To get the triangulation of the Borrmean link complement, 8 tetrahedra are needed.

2.2 Gluing

2.2.1 General structure of a tetrahedron

Our tetrahedron j , $j \in \{0, 1, 2, \dots, 7\}$ has four vertices that will be numbered from 0 to 3. In turn, it has six edges, to which we will assign the complex z_{jk} , $k \in \{1, 2, 3\}$. In general, the tetrahedron j will have three

pairs of complexes z_{j1}, z_{j2}, z_{j3} , where each z_{jk} will be on opposite sides concerning vertex 3. In Fig[??], you can see how the complexes will be assigned to each tetrahedron so they will have this structure.

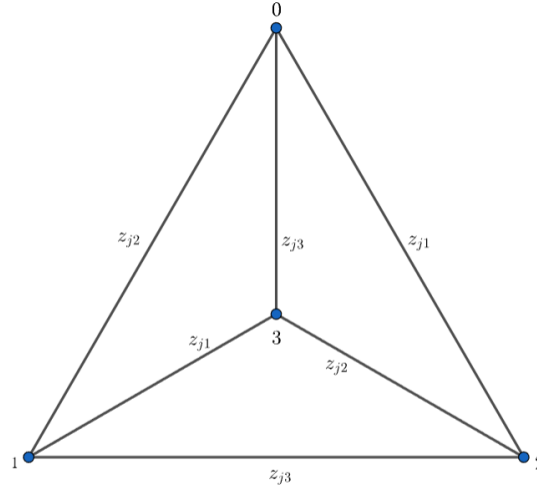


Figure 8: The general structure of a tetrahedron.

Since each z_{jk} is being assigned to an invariant axis, we will directly have the following system of equations:

$$z_{j1} = z_j \quad z_{j2} = \frac{1}{1 - z_j} \quad z_{j3} = \frac{z_j - 1}{z_j} \quad (6)$$

2.2.2 Faces

To triangulate the cusp of the Borromean link, we will use the face information to know which faces of one tetrahedron match with the faces of another tetrahedron.

Triangles (Borromean rings (Complement))			
Triangle #	Type	Degree	Tetrahedra (Tet vertices)
0	Triangle	2	0 (012), 2 (320)
1	Scarf	2	3 (213), 0 (013)
2	Triangle	2	0 (023), 1 (013)
3	Cone	2	4 (132), 0 (123)
4	Triangle	2	1 (012), 3 (203)
5	Cone	2	1 (023), 5 (032)
6	Scarf	2	2 (012), 1 (123)
7	Triangle	2	3 (012), 2 (013)
8	Cone	2	7 (132), 2 (123)
9	Cone	2	6 (012), 3 (013)
10	Scarf	2	4 (012), 6 (312)
11	Triangle	2	7 (230), 4 (013)
12	Triangle	2	4 (023), 5 (021)
13	Triangle	2	6 (302), 5 (013)
14	Scarf	2	7 (031), 5 (123)
15	Triangle	2	7 (012), 6 (013)

Figure 9: Equivalent faces in the eight tetrahedra necessary for the triangulation of the cusp of the Borromean bond.

Fig[??] shows the triangles divided into sixteen pairs so that each pair describes which triangles they are equivalent to and which tetrahedra they belong to. In turn, Fig[??] shows visually how the match between the faces results.

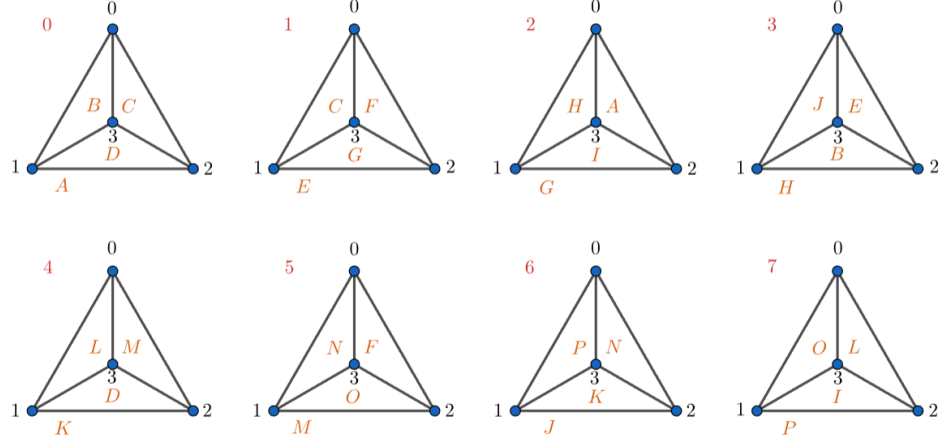


Figure 10: Visualization of the equivalent faces shown in Fig[??].

The tetrahedron number appears in red, and a letter has been assigned to each face that expresses the equivalent faces. Faces with the same letter are matching faces. The letter outside the tetrahedron points to the face that cannot be seen. All tetrahedra have the structure proposed in Fig[??].

2.2.3 Ticks

Despite knowing which faces stick together, this information is not enough; it is necessary to know which edges are equivalent at the time of gluing because the faces are joined has a specific configuration of action for this, Fig[??] is helpful to indicate which edges are equivalent to each other.

Edge #	Type	Degree	Tetrahedra (Tet vertices)
0		8	0 (01), 3 (21), 2 (31), 7 (21), 6 (31), 4 (01), 7 (23), 2 (32)
1		4	0 (02), 2 (30), 3 (20), 1 (01)
2		8	0 (03), 1 (03), 5 (02), 4 (02), 6 (32), 5 (03), 1 (02), 3 (23)
3		8	0 (12), 4 (13), 7 (30), 5 (21), 4 (23), 0 (32), 1 (31), 2 (20)
4		4	0 (13), 3 (13), 6 (12), 4 (12)
5		8	1 (12), 2 (01), 3 (01), 6 (01), 7 (01), 5 (13), 6 (02), 3 (03)
6		4	1 (23), 2 (12), 7 (13), 5 (32)
7		4	4 (03), 5 (01), 6 (30), 7 (20)

Figure 11: Equivalent edges for gluing equations and Borromean link cusp triangulation.

Visually Fig[??] shows how edges relate to others, with a tick that identifies each group of equivalent edges.

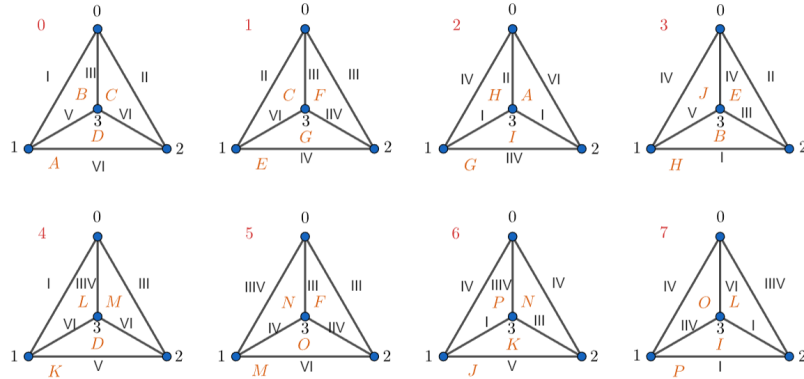


Figure 12: The visual shape of the equivalent edges shown in Fig[??].

The marks have been written in roman numerals. Having eight different marks, the Roman numerals from 1 to 8 will be seen on the edges. Al mismo tiempo, there will be eight gluing equations.

2.2.4 Orientations

Another piece of information that must be rescued to glue our faces properly is that we must recognize how the edges of our tetrahedra are oriented. From Fig[??] it is possible to obtain these orientations.

To give a proper orientation to the tetrahedra, I have first used a standard orientation for the 0 tetrad. With this, we check if the established orientation goes in the same direction as marked in Fig[?]. If the orientation goes in the same direction, all equivalent brands go in that same direction; On the other hand, if the orientation goes in the opposite direction to what Fig[?] marks, all the equivalent marks go in the opposite direction.

Once some tetrahedra are oriented, it will be visible that not all tetrahedra were completely oriented. To solve this, we take some tetrahedron that has not finished orienting itself and apply the standard form to the missing edges. Then, the equivalent edges are oriented and the process is repeated until everything is consistently oriented.

In Fig[?] you can see how the tetrahedra turn out after being oriented. An example of what is explained is, for example, the segment 0 (10), which means that for the tetrahedron 0, the vertex 1 travels to the vertex 0. In Fig[?] the directions appear in the opposite direction, therefore, all the edges equivalent to 0 (01) will be in the opposite direction. Now, if we see the segment 0 (02), for the tetrahedron 0, the vertex 0 travels to the vertex 2, being the same direction as mentioned in Fig[?] , therefore, all equivalent edges do not go in opposite directions.

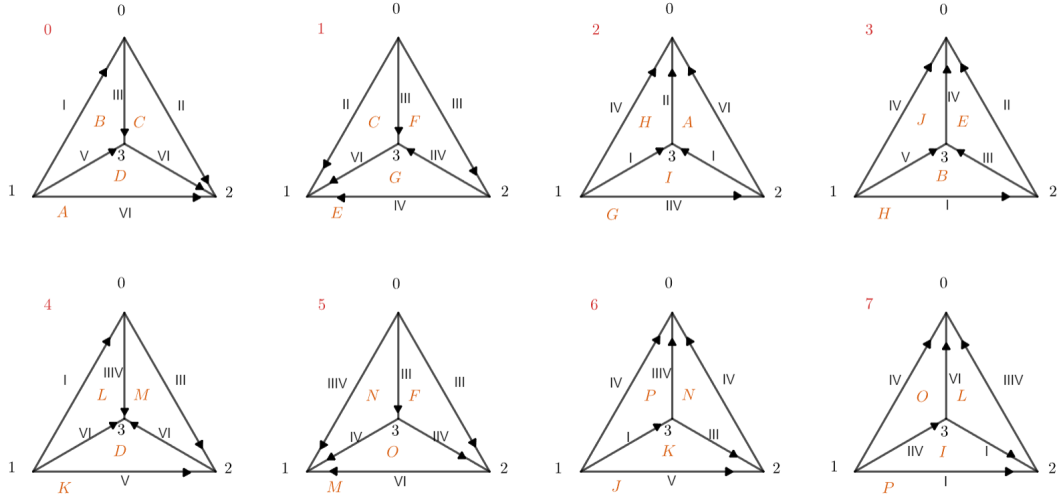


Figure 13: Orientations of the edges of each tetrahedron.

2.2.5 Components

As visible in Fig[6], the Borromean link has three components. This implies that to get the triangulation of our cusps, the three groups of equations will be required, each group for each component.

Fig[?] shows which vertices belong to each component, so we will have two components using eight vertices and one component that uses 16 vertices.

Vertices (Borromean rings (Complement))			
Vertex #	Type	Degree	Tetrahedra (Tet vertices)
0	Ideal: Torus	8	0 (0), 1 (0), 3 (2), 2 (3), 5 (0), 7 (2), 6 (3), 4 (0)
1	Ideal: Torus	16	0 (1), 4 (1), 3 (1), 2 (2), 7 (3), 6 (1), 2 (1), 1 (3), 5 (2), 7 (1), 1 (2), 0 (3), 4 (2), 5 (3), 3 (3), 6 (2)
2	Ideal: Torus	8	0 (2), 4 (3), 1 (1), 2 (0), 5 (1), 7 (0), 3 (0), 6 (0)

Figure 14: Vertices that belong to the same component of the triangulation of the cusp of the Borromean link.

2.3 Gluing equations

Using the edge invariants (equations 3, 4 and 5) and by knowing the polyhedron obtained from the gluing and using the knowledge from the ideal tetrahedron, the equation:

$$\prod z(e_i) = 1 \quad (7)$$

must satisfy in order for the last tetrahedron meet with the first at the face with vertices at 0, 1 and ∞ . Also, since there are no crossings between the tetrahedron, the equation:

$$\sum \arg(z(e_i)) = 2\pi \quad (8)$$

must satisfy.

Equations ?? and ?? are known as the *edge gluing equations*. These equations ensure that the gluing is a complete cycle and with no crossings or intersections between tetrahedra. The pair of these equations must satisfy for every edge class (every group of edges with the same amount of ticks) seperately.

To obtain the gluing equations, what we do is consider the edges that have the same mark. These edges are multiplied among themselves because they fulfill the role of being our edge invariants, the result of which is equivalent to 1. In turn, the sum of the arguments of the complexes used for each equation is identical to 2π .

2.4 Completeness equations

2.4.1 Cusp triangulations

The Borromean link has three components to form its cusp triangulations. For each component, an Euclidean structure must be generated allowing us to get a complete hyperbolic structure. From this, how the faces of our truncated tetrahedra are glued together is relevant, given that for each component, our tetrahedra will define which vertices of its tetrahedra will be found at the infinity of our hyperbolic space. Each component has a set of associated vertices, which are presented in Fig[??], where it's defined, for each component, which vertices will be found at infinity for its respective cusp triangulation. For each component, its triangulation is obtained when it is possible to pass through each vertex involved entirely. The way we glue the tetrahedra of our cusp triangulation is described in the gluing subsection.

When we generate our cusp triangulations, the relevance of them being presented as an Euclidean structure on the torus body lies in the fact that this implies that our cusp was generated from translations between the members of the component, without rotations or scaling. For each tetrahedron present in our triangulation, its location in the plane has been obtained by multiplying each vertex by the edges invariants obtained from previously arranged tetrahedra. After multiplying by all these associated complexes, a result equivalent to 1 should be obtained, implying that the translation was successful. Preserving this property is very important, otherwise, we cannot ensure the plane is completely covered.

2.4.2 Meridian and parallel selection

Because our components have a complete hyperbolic structure, this implies that the shape of our components must be a parallelogram, with two directions (a meridian and a parallel). These two directions follow the same idea of translations, therefore, when we travel through our component either through the meridian or through the parallel, the directions have an associated complex that must be equivalent to 1, which is called $H(\alpha)$ and $H(\beta)$. How these directions are traversed must be such that from opposite sides of our parallelograms, we get the same edge, its equivalent. The meridians and parallels chosen can be arbitrary; if they follow this idea of equivalent edges, we will obtain the same complexes $H(\alpha)$ and $H(\beta)$.

3 Results

3.1 Gluing equations

Knowing how gluing equations works and considering the structure of the tetrahedra that was mentioned in Fig[??], the equations obtained are the following:

$$z_{02}z_{21}z_{22}z_{33}z_{42}z_{61}z_{72}z_{73} = 1 \quad (9)$$

$$\text{Arg}(z_{02}) + \text{Arg}(z_{21}) + \text{Arg}(z_{22}) + \text{Arg}(z_{33}) + \text{Arg}(z_{42}) + \text{Arg}(z_{61}) + \text{Arg}(z_{72}) + \text{Arg}(z_{73}) = 2\pi$$

$$z_{01}z_{12}z_{23}z_{31} = 1 \quad (10)$$

$$\text{Arg}(z_{01}) + \text{Arg}(z_{12}) + \text{Arg}(z_{23}) + \text{Arg}(z_{31}) = 2\pi$$

$$z_{03}z_{11}z_{13}z_{32}z_{41}z_{51}z_{53}z_{62} = 1 \quad (11)$$

$$\text{Arg}(z_{03}) + \text{Arg}(z_{11}) + \text{Arg}(z_{13}) + \text{Arg}(z_{32}) + \text{Arg}(z_{41}) + \text{Arg}(z_{51}) + \text{Arg}(z_{53}) + \text{Arg}(z_{62}) = 2\pi$$

$$z_{02}z_{03}z_{11}z_{21}z_{41}z_{53}z_{73}z_{42} = 1 \quad (12)$$

$$\text{Arg}(z_{02}) + \text{Arg}(z_{03}) + \text{Arg}(z_{11}) + \text{Arg}(z_{21}) + \text{Arg}(z_{41}) + \text{Arg}(z_{53}) + \text{Arg}(z_{73}) + \text{Arg}(z_{42}) = 2\pi$$

$$z_{01}z_{31}z_{43}z_{63} = 1 \quad (13)$$

$$\text{Arg}(z_{01}) + \text{Arg}(z_{31}) + \text{Arg}(z_{43}) + \text{Arg}(z_{63}) = 2\pi$$

$$z_{13}z_{22}z_{32}z_{33}z_{51}z_{61}z_{62}z_{72} = 1 \quad (14)$$

$$\text{Arg}(z_{13}) + \text{Arg}(z_{22}) + \text{Arg}(z_{32}) + \text{Arg}(z_{33}) + \text{Arg}(z_{51}) + \text{Arg}(z_{61}) + \text{Arg}(z_{62}) + \text{Arg}(z_{72}) = 2\pi$$

$$z_{12}z_{23}z_{52}z_{71} = 1 \quad (15)$$

$$\text{Arg}(z_{12}) + \text{Arg}(z_{23}) + \text{Arg}(z_{52}) + \text{Arg}(z_{71}) = 2\pi$$

$$z_{43}z_{52}z_{63}z_{71} = 1 \quad (16)$$

$$\text{Arg}(z_{43}) + \text{Arg}(z_{52}) + \text{Arg}(z_{63}) + \text{Arg}(z_{71}) = 2\pi$$

3.2 Completeness equations

For each component, a path in green and another in orange have been chosen that goes from left to right and from bottom to top, respectively. Both paths have been formed so that the routes begin and end on equivalent faces. The route from left to right could have been more complex but, theoretically, regardless of the path followed the result should be the same, therefore, we took the easiest path. The same thing happens analogously for the route from bottom to top.

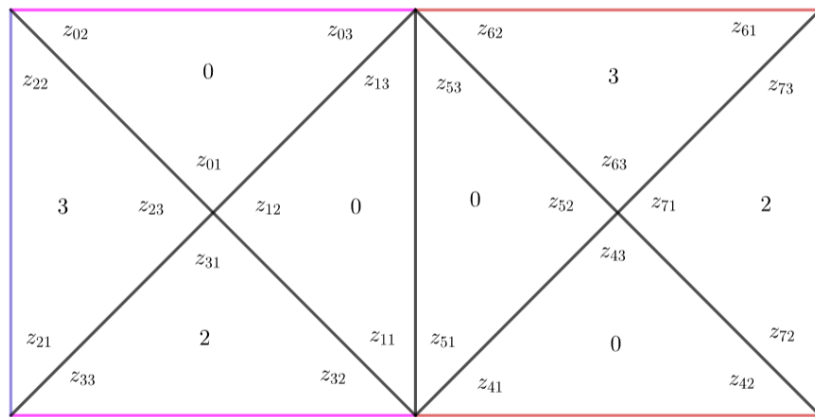


Figure 15: Triangulation of the 1st component.

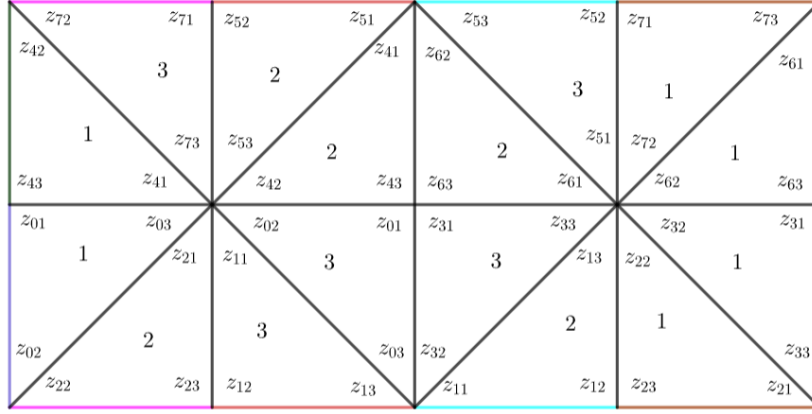


Figure 16: Triangulation of the 2nd component.

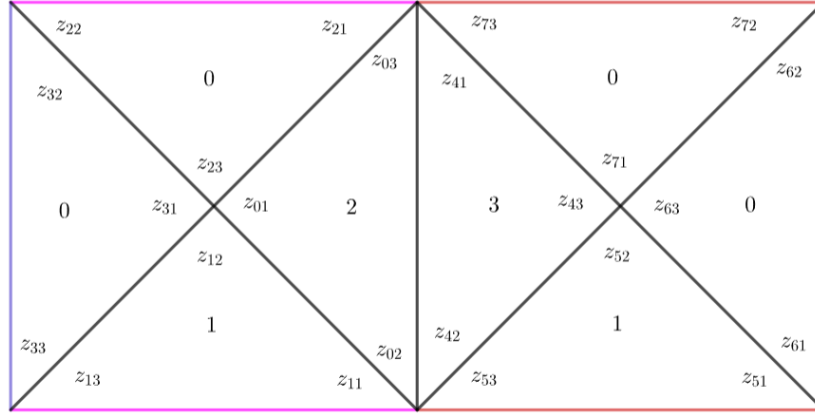


Figure 17: Triangulation of the 3rd component.

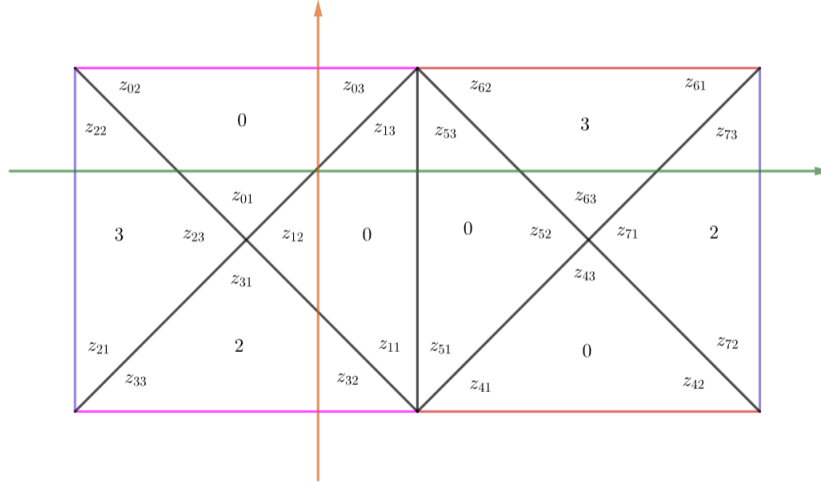


Figure 18: Meridian and parallel of the 1st component.

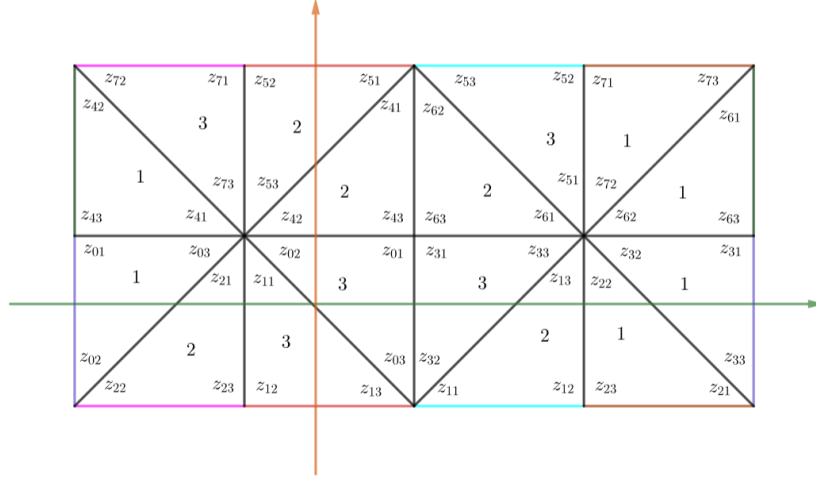


Figure 19: Meridian and parallel of the 2nd component.

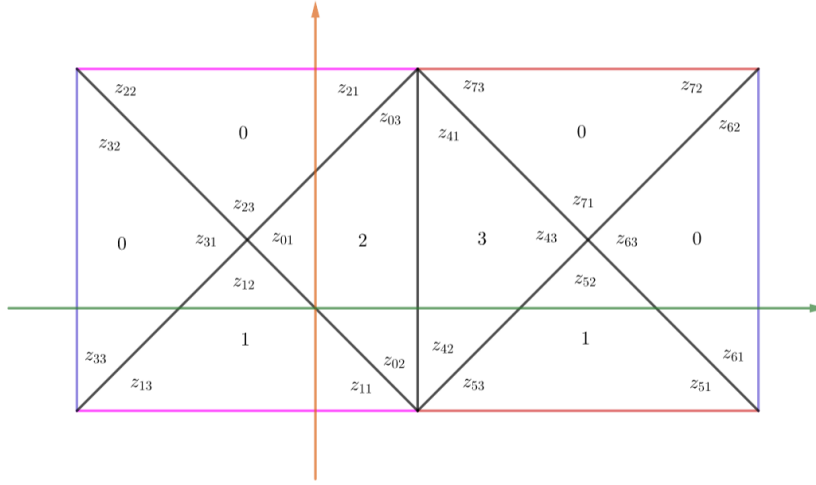


Figure 20: Meridian and parallel of the 3rd component.

For Fig[??] we will have the equations:

$$H(\alpha) = z_{22}z_{01}^{-1}z_{13}z_{53}z_{63}^{-1}z_{73} = 1 \quad H(\beta) = z_{32}^{-1}z_{12}z_{03}^{-1} = 1 \quad (17)$$

For Fig[??] we will have the equations:

$$H(\alpha) = z_{02}^{-1}z_{21}z_{11}z_{03}^{-1}z_{32}^{-1}z_{13}z_{22}z_{33}^{-1} = 1 \quad H(\beta) = z_{13}^{-1}z_{02}z_{42}z_{51}^{-1} = 1 \quad (18)$$

For Fig[??] we will have the equations:

$$H(\alpha) = z_{33}^{-1}z_{12}z_{02}^{-1}z_{42}^{-1}z_{52}z_{61}^{-1} = 1 \quad H(\beta) = z_{11}^{-1}z_{01}z_{21}^{-1} = 1 \quad (19)$$

4 Discussion

4.1 Gluing equations

Regarding the gluing equations, as previously mentioned, their derivation stems from the ideal tetrahedron. In the context of this article, examination of the proposer triangulation of every component (Fig[??], Fig[??], and Fig[??]) reveals that the summation of angles employed in the gluing equations must result in 2π , given the completion of a circle at the vertex joints. A comparison between the generated components and the gluing equations presented in Section ?? demonstrates the fulfillment of this criteria. Additionally, it is discerned that for the adequate adhesion of corresponding vertices, the edges must possess uniform dimensions; this assertion is substantiated by the scrutinization of the obtained gluing equation, thereby ensuring this outcome.

4.2 Completeness equations

In Fig[??] it can be seen that all the vertices associated with the first component are used. This triangulation is a generator of the first cusp, therefore, no matter which direction you add triangles, you will arrive at one that has already been placed, so that everything is consistently glued together.

Similarly, in Fig[??] it can be seen that all the vertices associated with the second component are used, and in Fig[??] You can see that all the vertices associated with the third component are used.

Note that in the triangulation of each component, at the intersections of several tetrahedra that form a “turn”, only the complexes that are involved in the same system of gluing equations share a corner. For clarity, for each component, the sides were marked with colors that identify which part of the component sticks to which part of the component. In turn, do not forget that for each complex z_{jk} , the subscript j indicates the tetrahedron used. For each tetrahedron j , the number that appears in the center of each tetrahedron seen from above represents the vertex that is at infinity.

In general, it can be stated that the triangulation proposal generated through this article is valid because the necessary requirements for filling out the desired plans are met. All faces of each of the component tetrahedra have been used, likewise the vertices are joined appropriately between faces using the equivalent vertices. In addition to this, it is observed that the parallelogram obtained is consistent, this means that to join it with another completely identical one, it would not be necessary to dilate, contract or rotate, this allows the use of our parallelograms to completely fill the plane.

5 References